The Impact of Hospital-Plan Vertical Integration on Health Outcomes

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PRELIMINARY

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Abstract

The healthcare sector has become increasingly vertically integrated, with hospital systems integrating healthcare financing by launching their own health plans. This paper investigates the causal effect of hospital health-plan vertical integration on health outcomes using data on Medicare Advantage beneficiaries. We distinguish three causal objects: (i) the effect on patient outcomes of being enrolled in a provider-sponsored MA plan (PSHP), (ii) the effect of receiving care at a vertically integrated (VI) hospital, and (iii) the interaction of the two. For the plan channel, we adopt an instrumental variables approach that that combines (a) the quasi-random differences in PSHPs availability within zip codes that span more than one county. For the hospital channel, we exploit an ambulance-based instrument for hospital assignment. We find that PSHP have lower readmission rate by 1.7% than standard plans but the difference fades as the hospital markets become more concentrated.

Keywords: Health Economics, Medicare Advantage, Vertical Integration, Provider-Sponsored Health Plan.

1 Introduction

Health care market consolidation in the United States has evolved over time. Historically, consolidation consisted primarily of hospitals mergers and insurers mergers, while more recently hospital and insurers are becoming increasingly vertically integrated. This latter form of integration is not only the result of mergers and acquisitions, but comes in the form of large health systems that integrate the insurance business with healthcare delivery. In recent years, there has been a rise in hospitals and health systems offering their own health plans. As of 2019, approximately 37 percent of Medicare beneficiaries are enrolled in a MA plan and 20 percent of MA plans are sponsored by a health system. See 1-4.

The literature on hospital-insurer vertical integration is primarily focused the trade-off between efficiency and market foreclosure. Instead, little is know about the effect of hospital sponsored health plans on healthcare utilization and health outcomes. Proponents argue that the vertical integration of providers and insurers can lead to administrative and clinical integration, possibly achieving high efficiency and quality (Orszag). However, opponents suspect that profit motives are the driving force behind vertical integration and believe that the vertical integration of providers and insurers may increase health care utilization and costs without necessarily improving quality (Geruso).

Medicare Advantage beneficiaries may be exposed to different level of vertical integration (VI) depending on the plan they are enrolled in and the provider from which they receive care. First, beneficiaries are not exposed to VI when they are on a non-VI plan and receive care from a non-VI provider. Second, some beneficiaries are on an integrated plan but admitted to a nonintegrated hospital. Third, some beneficiaries are on an non-VI plan and are admitted to an integrated hospital. Finally, some beneficiaries who experienced full integration when they are on a VI plan and receive care from the hospital that sponsors that plan.

The objective of this paper is to estimate the impact of insurer-provider vertical integration on healthcare utilization measures like hospital length of stay, diagnosis intensity, 30-day readmission and discharge location. In particular, we disentangle the differential impact due to the different channels of exposure to plan-provider integration:

1. a plan-side channel: the effect of enrolling in a provider-sponsored MA plan (PSHP),
2. a provider-side channel: the effect of receiving care at a hospital that is vertically integrated (VI) with a plan sponsor, and
3. a full-integration channel: a PSHP enrollee treated at their own plan's integrated hospital

We implement an empirical strategy to disentangle these channels. The novelty is to bring together two sources of quasi-experimental variation tailored to each margin. For the hospital channel, we adopt an ambulance-preference instrument that shifts emergency hospital assignment. For the plan channel, we exploit two sources of variation: first, we rely on the plausibly quasi-random differences in PSHPs availability across adjacent counties, and we use the shift in PSHPs attractiveness that is driven by the relative distance between the nearest non-VI hospital and the nearest VI hospital. For the full integration channel, we simply take the product of the hospital channel and the plan channel. The combined design delivers separate local average treatment effects (LATEs) for the plan-side and provider-side exposures and their interaction for emergency episodes. By separating plan and provider margins, our estimates inform whether observed differences arise from insurance design and network management (plan-side) or from practice patterns at integrated delivery sites (provider-side).

Relation to the literature. Our design builds on four literatures. First, a growing body of evidence compares integrated and non-integrated MA plans, documenting higher star ratings and premiums as well as lower non-claims (administrative) costs in integrated plans, with mixed equity implications (e.g., Park et al., 2023). Second, encounter-level comparisons in integrated contexts show differences in coding intensity, ICU use, mortality, and plan retention (e.g., Meyers et al., 2020). These associations motivate but do not identify causal effects. Third, the health economic theory of hospital insurer vertical integration emphasizes a trade-off between coordination benefits and upcoding incentives. In particular, Geruso and Layton (2020) shows that in the MS market, VI plans' coding intensity is substantially higher than that of non-VI plan. Fourth, the emergency-care identification strategy that exploits ambulance-company hospital preferences provides a credible way to exogenously shift *hospital choice* for acute cases (Doyle Jr et al., 2015). We adapt the latter to instrument provider-side exposure (VI hospital) while separately instrumenting plan-side exposure (PSHP enrollment) using differential distance.

Contribution First, we provide a unified framework to separate plan-side and provider-side VI effects and their interaction. Second, we adapt ambulance-preference IV to the VI context, enabling credible estimation for emergency episodes. Third, we connect encounter-level effects to plan-level features documented in the MA literature (premiums, non-claims costs, quality), clarifying the mechanisms by which integration may affect value.

Methodologically, we show how to jointly identify two endogenous exposures and their synergy at the episode level, tying plan design to site-of-care assignment. Substantively, focusing on emergencies ensures that plan networks and beneficiary preferences are least able to confound hospital assignment, isolating provider-side behavior at the moment of acute care while still allowing plan-side design to operate through post-acute pathways.

2 Illustrative Model

This section develops a simple static model to illustrate how different degrees of hospital–insurer vertical integration affect hospitals' incentives to invest in care coordination and, ultimately, readmission probabilities. The goal is not to provide a fully realistic structural model, but rather to generate sharp, testable predictions for how readmissions should differ across organizational scenarios that differ in how closely the enrollee's plan is linked to the treating hospital.

2.1 A Simple Model of Vertical Integration and Readmissions

Consider a single index admission for a patient enrolled in a Medicare Advantage (MA) plan and treated at a hospital. The hospital chooses effort $e \geq 0$ in care coordination (e.g., discharge planning, follow-up, medication reconciliation). Effort reduces the probability of readmission:

$$p(e) = \bar{p} - e, \qquad e \in [0, \bar{p}],$$

with $\bar{p} \in (0, 1)$ and $p'(e) = -1 < 0$. A readmission is financially undesirable for both sides: let $\mu_H = \text{DRG payment } (\omega) - \text{IP total cost } (\gamma) < 0$ denote the hospital's net margin on a readmission, and $\mu_P = \text{benchmark}(b) - \omega$ the plan's net margin impact of a readmission (given a fixed benchmark b). Effort is costly for the hospital, with

$$C(e) = \frac{c}{2}e^2, \qquad c > 0.$$

Vertical integration affects how much of the plan's savings from avoided readmissions the hospital internalizes when choosing e. I capture this with a scalar

$$\theta \in [0, 1],$$

interpreted as the share of the plan's readmission margin μ_P that enters the hospital's objective for this particular plan–hospital pair. The hospital's payoff is

$$\Pi_H(e; \theta) = p(e)\big(\mu_H + \theta\mu_P\big) - \frac{c}{2}e^2 = (\bar{p} - e)\big(\mu_H + \theta\mu_P\big) - \frac{c}{2}e^2.$$

The first-order condition is

$$\frac{d\Pi_H}{de} = -\big(\mu_H + \theta\mu_P\big) - ce = 0 \quad \Rightarrow \quad e(\theta) = -\frac{\mu_H + \theta\mu_P}{c}.$$

Hence, effort is strictly increasing in the degree of integration:

$$\frac{de(\theta)}{d\theta} = -\frac{\mu_P}{c} > 0.$$

The returns for reducing readmission are higher under vertical integration. The induced readmission probability is

$$p(\theta) = p\big(e(\theta)\big) = \bar{p} - e(\theta) = \bar{p} + \frac{\mu_H + \theta\mu_P}{c},$$

so $p(\theta)$ is strictly decreasing in θ. A non-integrated relationship corresponds to $\theta = 0$, and full internalization to $\theta = 1$.

To link this general setup to hospital-led integration, consider a system S that owns both a hospital H^S and a sponsored MA plan P^S. For a given patient episode, what changes across organizational configurations is whether the treating hospital is H^S or a non-integrated hospital H^0, whether the patient's plan is P^S or a non-integrated plan P^0, and whether the plan and hospital belong to the same sponsoring system.

Integration creates two conceptually distinct channels through which the hospital internalizes plan-side savings:

- A *hospital-side channel*: as part of S, the system hospital H^S adopts clinical protocols (discharge pathways, early follow-up, etc.) that reduce readmissions for all patients treated there, regardless of plan. This raises the effective θ whenever the treating hospital is vertically integrated.

- A *plan-side channel*: the system plan P^S invests in care management (post-discharge calls, telehealth check-ins, pharmacy support) that reduce readmissions for its enrollees even at non-system hospitals. This raises the effective θ whenever the enrollee is in a vertically integrated plan.

I model these channels parsimoniously by assigning, for each plan–hospital pair, an effective degree of internalization:

$$\theta = \begin{cases} 0, & \text{Non-VI plan, Non-VI hospital,} \\ \theta_P, & \text{VI plan, Non-VI hospital,} \\ \theta_H, & \text{Non-VI plan, VI hospital,} \\ \theta_H + \theta_P, & \text{VI plan, VI hospital in } \textit{different} \text{ systems,} \\ 1, & \text{VI plan, VI hospital in the } \textit{same} \text{ system.} \end{cases}$$

Here $\theta_P \in (0, 1)$ captures the incremental internalization associated with being in a VI plan (plan-side care management), and $\theta_H \in (0, 1)$ the incremental internalization from being treated at a VI hospital (clinical protocols). Consistent with the idea that clinical decisions are the main lever on readmissions, I assume

$$0 < \theta_P < \theta_H, \qquad \theta_H + \theta_P < 1.$$

The last line normalizes full internalization to $\theta = 1$ when the plan and hospital belong to the same sponsoring system; this is strictly stronger than the case where both sides are VI but belong to different systems.

This yields five organizational configurations for a given admission:

(i) **Non-VI plan, Non-VI hospital**: P^0–H^0. The episode is entirely outside any integrated system:

$$\theta_i = 0, \qquad e_i = -\frac{\mu_H}{c}, \qquad p_i = \bar{p} + \frac{\mu_H}{c}.$$

(ii) **VI plan, Non-VI hospital**: P^S–H^0. The enrollee is in a VI plan but treated at a Non-VI hospital. Plan-side care management applies, but hospital workflows are unchanged:

$$\theta_{ii} = \theta_P, \qquad e_{ii} = -\frac{\mu_H + \theta_P \mu_P}{c}, \qquad p_{ii} = \bar{p} + \frac{\mu_H + \theta_P \mu_P}{c}.$$

(iii) **Non-VI plan, VI hospital**: P^0–H^S. The enrollee is in a standard plan but treated at a VI hospital. System-wide clinical protocols apply to all patients of H^S:

$$\theta_{iii} = \theta_H, \qquad e_{iii} = -\frac{\mu_H + \theta_H \mu_P}{c}, \qquad p_{iii} = \bar{p} + \frac{\mu_H + \theta_H \mu_P}{c}.$$

(iv) **VI plan, VI hospital (different sponsors)**: P^{S_1}–H^{S_2} with $S_1 \neq S_2$. Both sides are

VI, but not with each other. The enrollee benefits from plan-side care management and hospital-side protocols, but the hospital does not fully internalize this particular plan's margin:

$$\theta_{iv} = \theta_H + \theta_P, \qquad e_{iv} = -\frac{\mu_H + (\theta_H + \theta_P)\mu_P}{c}, \qquad p_{iv} = \bar{p} + \frac{\mu_H + (\theta_H + \theta_P)\mu_P}{c}.$$

(v) **Fully integrated dyad (same sponsor)**: P^S–H^S. The plan and hospital are owned by the same system, and the hospital fully internalizes this plan's savings from avoided readmissions:

$$\theta_v = 1, \qquad e_v = -\frac{\mu_H + \mu_P}{c}, \qquad p_v = \bar{p} + \frac{\mu_H + \mu_P}{c}.$$

By construction,

$$0 = \theta_i < \theta_{ii} = \theta_P < \theta_{iii} = \theta_H < \theta_{iv} = \theta_H + \theta_P < \theta_v = 1,$$

so monotonicity of $e(\theta)$ and $p(\theta)$ immediately implies

$$e_v > e_{iv} > e_{iii} > e_{ii} > e_i \quad \Longleftrightarrow \quad p_v < p_{iv} < p_{iii} < p_{ii} < p_i.$$

Intuition. In the non-integrated case (i), the hospital only internalizes its own loss from readmissions and chooses the baseline effort e_i. In case (ii), enrollment in a VI plan activates plan-side care management that reduces readmissions even at a non-integrated hospital, so $e_{ii} > e_i$ and $p_{ii} < p_i$. In case (iii), being treated at a VI hospital exposes the patient to system-wide clinical protocols; these clinical levers are stronger than plan-side programs alone, so $e_{iii} > e_{ii}$ and $p_{iii} < p_{ii}$. In case (iv), the enrollee is in a VI plan and treated at a VI hospital, but the two belong to different systems; plan-side and hospital-side efforts both operate, but the hospital still does not fully internalize this plan's margin. Finally, in case (v), the patient is both in the system's plan and treated at the system's hospital, so clinical and care-management levers are fully aligned within the same organization; effort is highest and readmissions lowest.

2.2 Testable Implications

The toy model yields a simple set of comparative-static predictions that can discipline the empirical analysis. Let p_{kh} denote the risk-adjusted readmission probability for enrollees in plan type $k \in \{\text{Non-VI}, \text{VI}\}$ treated at hospital type $h \in \{\text{Non-VI}, \text{VI}\}$. The five configurations above correspond to:

- $p_{\text{Non-VI plan,Non-VI hosp}} = p_i$,
- $p_{\text{VI plan,Non-VI hosp}} = p_{ii}$,
- $p_{\text{Non-VI plan,VI hosp}} = p_{iii}$,
- $p_{\text{VI plan,VI hosp, diff system}} = p_{iv}$,
- $p_{\text{VI plan,VI hosp, same system}} = p_v$.

Under the assumptions above, the model implies the ordered inequalities

$$p_{\text{Non-VI plan,Non-VI hosp}} > p_{\text{VI plan,Non-VI hosp}} > p_{\text{Non-VI plan,VI hosp}} > p_{\text{VI plan,VI hosp, diff system}} > p_{\text{VI plan,VI hosp, same s}} \tag{1}$$

These inequalities provide a clear empirical ranking to test. After controlling for patient risk, case mix, and market conditions, the model predicts that:

- readmissions should be highest when neither the plan nor the hospital is vertically integrated;
- readmissions should be lower for enrollees of vertically integrated plans even when treated at non-integrated hospitals;
- readmissions should be lower still when non-integrated plans' enrollees are treated at vertically integrated hospitals, reflecting the greater importance of hospital-side clinical protocols;
- among episodes involving both a VI plan and a VI hospital, readmissions should be lower when the plan and hospital share the same sponsor than when they belong to different systems.

The empirical analysis can then examine whether the data are consistent with this predicted ordering, and to what extent departures from these inequalities inform the role of vertical integration in shaping incentives for care coordination and readmission reduction.

3 Motivating Evidence

A central premise of the paper is that hospital–insurer vertical integration can reduce readmissions by strengthening incentives and capabilities for care coordination. In the illustrative model, greater internalization of the plan-side margin leads hospitals to choose higher care-coordination effort and therefore lower readmission risk. Before turning to the instrumental variables design, this section presents suggestive (non-causal) evidence consistent with that mechanism and highlights an additional empirical pattern: the association between vertical integration and readmissions is strongest in competitive markets and attenuates as local hospital markets become more concentrated.

Visual evidence. Figure 1 provides a transparent visualization of this relationship. The figure residualizes readmission rates on the full set of controls and year/episode fixed effects, then plots binned means against HSA-level HHI separately for VI and non-VI hospitals, along with within-group linear fits. The VI series starts below the non-VI series in competitive HSAs, but rises more steeply with HHI, so the vertical gap narrows in highly concentrated HSAs.

Descriptive Evidence. We study how the within-hospital association between vertical integration and risk-adjusted readmissions varies with hospital market concentration. Let m index the hospital's local market (HSA in the main specification). We estimate:

$$\mathrm{RR}_{het} = \beta_1 \,\mathrm{VIHosp}_{ht} + \beta_2 \,\ln(\mathrm{HHI})_{mt} + \beta_3 \,\mathrm{VIHosp}_{ht} \times \ln(\mathrm{HHI})_{mt} + X'_{het}\gamma + \delta_e + \tau_t + \varepsilon_{het}, \tag{2}$$

where RR_{het} is the risk-adjusted 30-day readmission rate for hospital h, episode type e, and year t; VIHosp_{ht} indicates whether the admitting hospital is vertically integrated with a plan sponsor. We consider due market definitions, Hospital Service Area (HSA) and County Area, and compute the corresponding log Herfindahl–Hirschman Index ($\ln(\mathrm{HHI})_{mt}$) using acute-care bed shares. In the estimation $\ln(\mathrm{HHI})_{mt}$ is mean-centered to ease interpretation. We include a set of risk adjusters and control for a rich set of hospital and market characteristics, along with episode fixed effects δ_e and year fixed effects τ_t. Regressions are weighted by ln(discharges) and standard errors are clustered at the hospital (CCN) level.

Table 1 summarizes the key coefficients. Two facts stand out. First, at the average level of concentration (i.e., when $\ln(\mathrm{HHI})_{mt} = 0$), vertically integrated hospitals have lower readmission rates than otherwise similar non-integrated hospitals ($\beta_1 < 0$). Second, the interaction term is positive and statistically significant ($\beta_3 > 0$), implying that the VI–non-

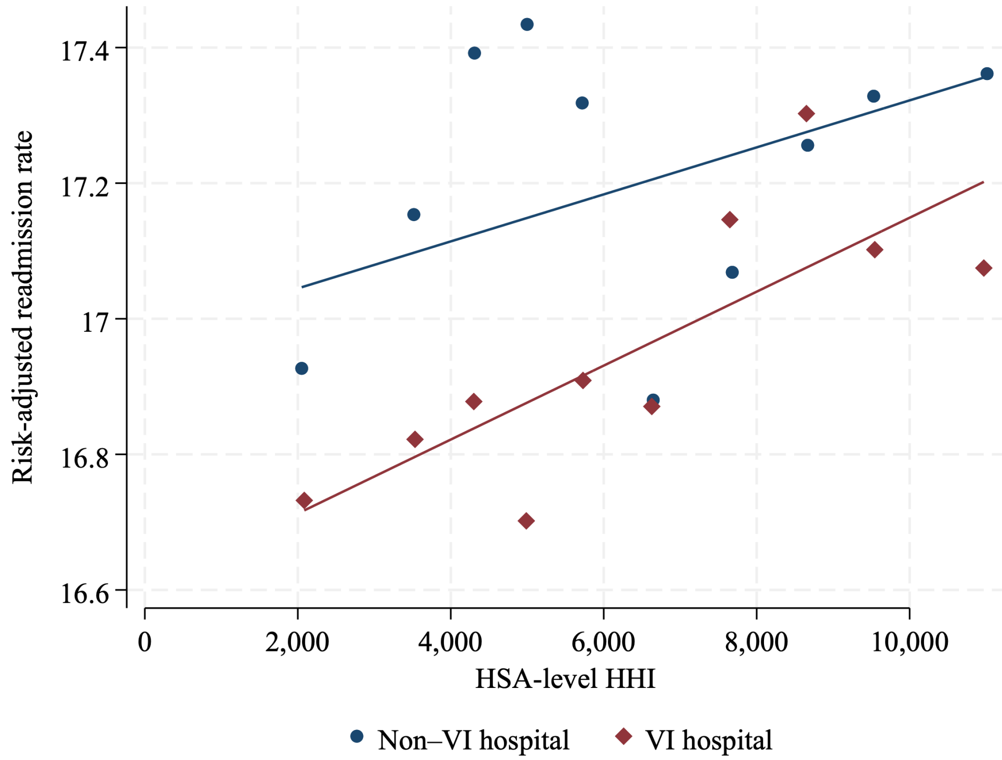

Figure 1: Readmission Rate vs. HHI by VI Hospital Status

Notes: Each point shows the mean readmission rate within each decile of the HSA-level HHI distribution, separately for VI and non-VI hospitals. The plotted outcome is residualized on a rich set of hospital and market controls, year fixed effects, and episode indicators, using ln(discharges) as analytic weights. The HSA-level HHI is constructed from acute-care bed shares. Lines are within-group linear fits.

VI readmission gap becomes smaller as hospital markets become more concentrated.

At the sample mean level of market concentration, vertically integrated (VI) hospitals are associated with *lower* risk-adjusted readmission rates. Because ln(HHI) is mean-centered, the coefficient on **VI hospital** captures the VI–non-VI difference evaluated at the **average** HHI in the estimation sample: in column (1) (HSA markets) the estimate -0.297 implies that VI hospitals have readmission rates that are 0.297 percentage points lower than non-VI hospitals at mean HHI; the analogous estimate in column (2) (county markets) is -0.288 percentage points. The positive and statistically significant interaction term, **VI hospital** $\times$ ln(**HHI**), implies that this VI advantage *shrinks* as market concentration increases. Using the point estimates, a 10% increase in HHI attenuates the VI–non-VI readmission gap by approximately 0.034 pp in HSA market and

Using the point estimates, a 10% increase in HHI relative to the mean reduces the magnitude of the VI effect by about $0.355 \times \ln(1.10) \approx 0.034$ in column (1) and $0.297 \times \ln(1.10) \approx$

0.028 in column (2). Interpreting these in percentage terms, this corresponds to an attenuation of roughly $100 \times (\exp(0.034) - 1) \approx 3.5\%$ (HSA) and $100 \times (\exp(0.028) - 1) \approx 2.9\%$ (county) in the VI–non-VI readmission gap.

Therefore, the main pattern we observe is that VI-hospitals are associated with lower level of risk adjusted readmission rate but this association fades away as the level of market concentration increases, and vanished in highly concentrated markets. However, we are not interpreting these results are causal since hospital integration status is endogenous, and unobserved differences across integrated and non-integrated hospitals (or across markets) could drive the observed gradient. This motivates the instrumental-variables design in the next section, which isolates plausibly exogenous variation in admission to a VI hospital and enrollment in PSHP.

Table 1: Readmission rates, vertical integration, and market concentration

	(1) Market = HSA	(2) Market = County
VI hospital	-0.297^{***}	-0.288^{***}
	(0.105)	(0.104)
ln(HHI)	-0.135	-0.111
	(0.190)	(0.178)
VI hospital × ln(HHI)	0.355^{***}	0.297^{***}
	(0.135)	(0.114)
HS hospital	-0.365^{***}	-0.378^{***}
	(0.099)	(0.099)
HS hospital × ln(HHI)	0.034	0.155
	(0.140)	(0.105)
ln(Expected readmission rate)	14.879^{***}	15.045^{***}
	(0.601)	(0.599)
ln(Risk score)	2.636^{***}	2.790^{***}
	(0.460)	(0.463)
Episode FE	Yes	Yes
Year FE	Yes	Yes
Weights	ln(discharges)	ln(discharges)
Clusters (CCN)	1,941	1,941
Observations	85,610	75,679

Notes: Unit of observation is hospital × episode type × year. The dependent variable is the risk-adjusted 30-day readmission rate. ln(HHI) is the log Herfindahl–Hirschman Index constructed from acute-care bed shares and mean-centered in the estimation sample. Markets are defined as HSAs in column (1) and counties in column (2). Both specifications include episode and year fixed effects and analytic weights equal to ln(discharges). A rich set of hospital and market-level controls are included but not reported. Standard errors (in parentheses) are clustered at the hospital (CCN) level. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.10$.

4 Institutional Setting

4.1 Medicare Advantage and provider-sponsored plans

Medicare Advantage (MA) is the alternative to traditional fee-for-service Medicare and covers its beneficiaries through contracts with private plans. Integration between the provider and the health plan is becoming a prominent feature of the Medicare Advantage (MA) program. As MA plans are paid on a capitated basis and receive additional bonus payments based on quality performance, there may be an increased incentive to provide integrated care across care settings in order to maintain healthcare cost below the capitated amount and improve their enrollees' health to achieve the quality-based bonus.

Provider-sponsored health plans (PSHPs) are vertically integrated entities where a provider organization owns and operates its own insurance plan to provide healthcare services. This integration combines the payer-side (insurance) and provider-side (hospitals, doctors) functions within a single organization. This organization often takes the form of a health system, but there are instances of health plans that are sponsored by provider organizations that do not include hospitals.

Medicare Advantage (MA) plans have specific service areas composed of set of nearby counties. MA beneficiaries may enroll only in plans filed for their county of residence. As shown in Figure 3, there is a lot of variation in the number of PSHPs' enrollment share in nearby counties. In particular, we rely on the fact that there are several zip codes whose area span more than one county. This scenario is illustrated in Figure 1. MA beneficiaries who live on the upper side of the border (yellow area) have the option to enroll in a PSHP, whereas beneficiaries who live in the lower side of the border do not have that option. Thus, two otherwise similar neighborhoods that straddle a county line can face different choice sets in terms of PSHP availability. This plausibly exogenous variation in the PSHPs availability within zip codes that span more than one county is a essential for our identification strategy.

4.2 Emergency care and ambulance assignment

The locus of treatment for emergency hospitalizations is, to a large extent, determined by pre-hospital factors, including ambulance transport decisions and patient location. Moreover, there is wide evidence that large city areas are often served by multiple ambulance companies, whereas smaller rural with a single ambulance company, neighboring companies provide service when the principal ambulance units are busy under so-called "mutual aid" agreements. Thus, in both cases, the assignment of the ambulance company to the patient

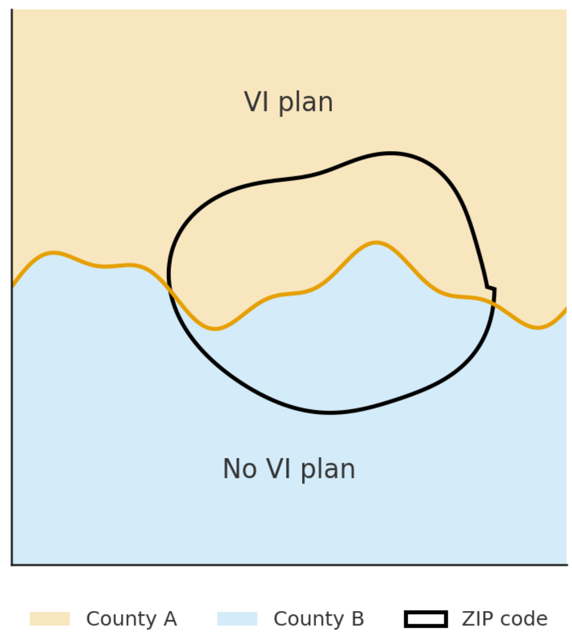

Figure 2: Within zip code variation in PSHP availability

is effectively random. (Doyle 2015, Chiang, et al., 2006 Ragone 2012).

In addition to plausibly exogenous assignment, ambulance companies are expected to have preferences for particular hospitals. In survey work described in Doyle et al. (2015), we found that paramedics have developed relationships with local emergency departments. For example, Skura (2001) studied ambulance assignment in the wake of a new system of competition between public and private ambulances in New York City. He found that patients living in the same ZIP code as public Health and Hospital Corporation hospitals were less than half as likely to be taken there when assigned a private, non-profit ambulance (29%) compared to when the dispatch system assigned them to an FDNY ambulance (64%). In most cases, the private ambulances were operated by non-profit hospitals and stationed near or even within those facilities, so they tended to take their patients to their affiliated hospitals. More broadly, with the exception of acute trauma care for which there are often defined local protocols for hospital assignment (Kahn et al. 2008) transport assignment for other emergencies is more likely to be driven by idiosyncratic preferences. Thus, in this context patient transport decisions are more likely to be made in ways that are not systematically tied to the underlying health of the patient.

Figure 2 illustrates the key source of quasi-random variation in hospital assignment that underlies the empirical strategy. When a medical emergency occurs, the dispatch system typically assigns a patient to the nearest available ambulance, which is effectively random from the patient's perspective. Conditional on the patient's location, the ambulance dispatched

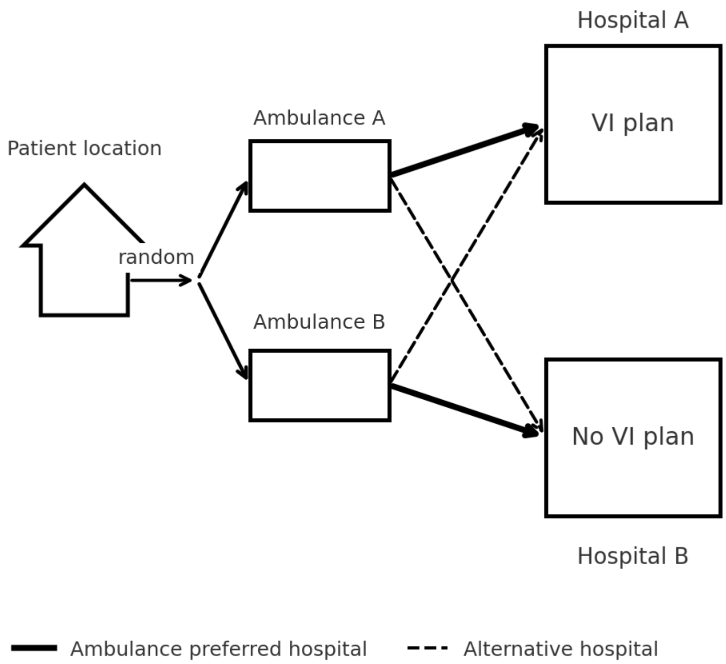

Figure 3: Hospital Assignment and Ambulance Preferences

may have systematically different "preferred" hospitals due to historical routing patterns, contractual relationships, or local familiarity.

As shown in Figure 2, two ambulances (A and B) serve the same geographic area but exhibit different propensities to transport patients to different hospitals. Ambulance A tends to deliver patients to Hospital A, which is vertically integrated (VI) with a Medicare Advantage plan, whereas Ambulance B tends to deliver patients to Hospital B, which is not integrated. Importantly, ambulances occasionally transport patients to the alternative hospital as well, generating natural variation in destination choice.

This setup creates a quasi-experimental shock in exposure to VI hospitals: patients who happen to be picked up by an ambulance with stronger ties to the VI hospital are more likely to be admitted there, even though the ambulance choice is conditionally random. This type of variation has been widely used in the hospital choice and emergency-care literature because it mimics an instrumental variable: ambulance identity predicts hospital assignment but is plausibly orthogonal to patient unobservables.

In this paper, I leverage this setting to isolate the causal effects of treatment within VI systems. Conceptually, the identification compares patients who live in the same area and experience similar emergencies, but who—due to the effectively random dispatch of a given ambulance—are differentially likely to be routed to a vertically integrated versus non-integrated hospital system.

In other words, patients typically do not choose the ambulance, and dispatch/rotation protocols and ambulance company service areas generate quasi-random assignment to companies with heterogeneous hospital destination preferences. This yields plausibly exogenous variation in the admitting hospital conditional on residential location and origin, forming the basis for ambulance preference instruments (Doyle Jr et al., 2015).

4.3 Outcome: 30-day unplanned hospital readmission

This paper focuses on hospital readmissions as a clinically meaningful and policy-relevant measure of care coordination and discharge quality. Readmissions are financially consequential for Medicare and hospitals, and they are explicitly targeted by Medicare's Hospital Readmissions Reduction Program (HRRP). Recent evidence indicates that a substantial share of acute admissions are followed by a rehospitalization within 30 days, with aggregate spending associated with readmissions on the order of tens of billions of dollars annually in Medicare.[1]

Definition and construction. Our main outcome, RR_{it}, is an indicator equal to one if beneficiary i experiences an *unplanned* acute inpatient readmission within 30 days of discharge from the *index* emergency admission at time t, and zero otherwise. We treat multiple rehospitalizations within the 30-day window as a single event, consistent with HRRP-style targeted measures, which are most naturally interpreted as readmission probabilities rather than counts.[2] Following CMS-style measurement conventions, we exclude same-day transfers and implement a planned-readmission algorithm to remove rehospitalizations that are predictably scheduled (e.g., selected elective procedures) rather than reflecting failures of post-discharge management.[3] Because our data are Medicare Advantage encounters, we can track readmissions to *any* acute-care hospital, not only the index hospital.

All specifications adjust for rich clinical risk at the patient level. In addition to demographics and a baseline risk score, we include Hierarchical Condition Category (HCC) comorbidity indicators and principal-diagnosis fixed effects for the index admission. This aligns with the spirit of CMS risk adjustment, which uses case-level models with extensive risk factors when computing risk-adjusted readmission measures.[4]

[1] See, e.g., **?** for summary statistics and policy discussion.

[2] **?** notes that CMS does not differentiate between one and multiple readmissions within the 30-day period when constructing targeted readmission metrics.

[3] **?** describe the 30-day all-cause framework and common exclusions used in HRRP-style definitions.

[4] **?** describes CMS's use of hierarchical logistic regression with a large set of patient risk factors for condition-specific readmission measures.

Why readmissions are informative for this setting. The policy motivation for using readmissions is that a non-trivial share of rehospitalizations may be avoidable through improved care transitions and coordination. In our context, this maps naturally to organizational features that affect discharge planning, follow-up management, and the integration of inpatient care with post-acute and outpatient settings.

Financial incentives and the HRRP. HRRP, implemented beginning in 2013, links hospital revenue to readmission performance. CMS evaluates hospital readmission performance over a multi-year window and applies payment reductions to hospitals with "excess" readmissions relative to national benchmarks.[5] The program initially targeted acute myocardial infarction, heart failure, and pneumonia, and later expanded to additional conditions and procedures.[6] Penalties were phased in from a 1% cap in 2013 to 2% in 2014 and 3% from 2015 onward.[7] Thus, readmissions affect hospital financial performance both directly (through costly utilization) and indirectly through explicit Medicare payment reductions tied to risk-adjusted readmission rates.[8]

Connection to vertical integration and this paper's contribution. A central hypothesis of this paper is that hospital–insurer vertical integration changes both *incentives* and *capabilities* to reduce readmissions. Provider-sponsored plans internalize savings from avoided utilization and can invest in care management and care coordination infrastructure; hospitals that sponsor plans may have stronger incentives to reduce avoidable rehospitalizations. When the plan and hospital share a common sponsor (full integration), incentives may be aligned most tightly.

Relative to the HRRP literature, which primarily studies hospitals in traditional Medicare and the average effects of pay-for-performance incentives, this paper examines how *organizational form* in Medicare Advantage—plan integration, hospital integration, and full integration—shapes readmission outcomes. Methodologically, we combine quasi-exogenous variation in emergency hospital assignment (ambulance preference) with border-driven variation in PSHP availability and geographic differential distance instruments to identify causal integration effects. Substantively, we connect integration effects to market structure by testing whether the readmission advantage of integrated hospitals attenuates as hospital markets become more concentrated, consistent with the descriptive evidence in Section 3.

[5] **?** provides a detailed description of the HRRP penalty structure, including the use of risk-adjusted readmission rates over a three-year window and benchmarking against the national mean.

[6] **?** summarizes the targeted conditions/procedures and the program timeline.

[7] **?**.

[8] **?** discusses how excess readmission rates translate into payment reductions under HRRP.

5 Data and Descriptive Evidence

Provider-Sponsored Health Plans To identify integrated MA plans sponsored by health systems, we used the AHRQ Compendium of Health Systems which contains the list of all health pans offered by hospitals and health systems. We then linked all of these hospitals to their CMS provider identification numbers (CCN) to identify them in AHA database and add information on hospital characteristics. To identify health plans sponsored by other provider organizations, we searched through the full list of MA contracts to flag those whose sponsor is a provider organization.

Table 1 shows some descriptive statistics on the Enrollment Share of PSHP across counties. Each county is weighed by the number of eligible beneficiaries. Panel A shows the distribution among counties where MA penetration rate is below 50%. Among these counties, the PSHP share is almost zero in more than half the counties. The mean is significantly larger than the median which implies that there is large variation across counties. Panel B shows the distribution among counties where MA penetration rate in above 50%. We observe that PSHPs plans' enrollment share is much larger among HMO plans. For the latter, the mean and the median are similar and there is less variation compared to the other set of counties.

Table 2: Descriptive Statistics of County PSHP Availability

Plan type	**Median**	**Mean**	**SD**	**Within-ZIP SD**
Panel A: Counties with MA penetration $\leq 50\%$				
HMO	0.029	0.149	0.509	32.7%
PPO	0.004	0.049	0.202	30.6%
Total	0.010	0.098	0.390	31.4%
Panel B: Counties with MA penetration $> 50\%$				
HMO	0.027	0.060	0.145	17.5%
PPO	0.003	0.020	0.080	14.6%
Total	0.016	0.040	0.119	15.8%

Notes: Table reports the first quartile, median, mean, standard deviation, and third quartile of county-level VI plan enrollment shares, separately for HMO and PPO plans. Statistics are weighted by the number of eligible beneficiaries in each plan type at the county level. Panels split counties by MA penetration at county level: Panel A includes counties with penetration below 50%, and Panel B counties with penetration above 50%. The column "Within-ZIP share" is computed as $1 - R^2$ from a eligible-weighted regression of the county-level VI-plan measure on ZIP fixed effects (sample restricted as indicated). This is the fraction of the observed, county-eligible-weighted variation attributable to differences within the same ZIP across counties.

Figures 3 and 4 in Appendix A complement these summary statistics by mapping the county-level number of PSHP per thousands of beneficiaries enrolled in MA. Figure 3 (Figure

4) shows the number of HMO (PPO) plans sponsored by health system per thousands of beneficiaries enrolled in HMO (PPO) plans in each county. Vertically integrated plans are highly concentrated in specific regions—such as California, the Pacific Northwest, the Upper Midwest, and parts of the Southwest—while many neighboring counties exhibit little to no PSHP presence. In particular, the figure reveals sharp discontinuities in PSHP availability between adjacent counties. This stark cross–county heterogeneity, often occurring over very short geographic distances, is precisely the spatial variation that underlies our identification strategy, as beneficiaries living in the same ZIP code but assigned to different counties face meaningfully different exposure to PSHP availability.

Medicare Advantage Claims Data I use administrative data from the Centers for Medicare & Medicaid Services (CMS) for the years 2019 and 2022. The main data source for plan enrollment is the Master Beneficiary Summary File (MBSF), which contains individual-level data on a 5% random sample of Medicare beneficiaries. The MBSF records enrollment information for each beneficiary, including whether they are enrolled in Traditional Medicare or a Medicare Advantage (MA) plan. The MBSF also includes detailed beneficiary characteristics such as age, gender, race, income measures, and county of residence. In 2019, total Medicare enrollment was approximately 61 million, so the 5% sample includes about 3 million individuals. Roughly 37% of beneficiaries were enrolled in MA plans. I exclude ESRD patients from the analysis and also special needs plans, pace plans, employer-sponsored plans or PFFS plans which have a different reimbursement scheme from CMS which is different from the standard plans. The main data source for the patients' health claims is the Inpatient Encounter Data, from which we can obtain the ambulance company and the hospital identifiers where patients were treated, the ICD-10 diagnosis codes, the procedure codes, the length of hospital stay, and the discharge location. In addition, I use the Inpatient, Outpatient, Skilled Nursing Facility (SNF), Home Health Agency (HHA), Durable Medical Equipment (DME), and Carrier Encounter files to calculate the average cost of care for Medicare beneficiaries.

Sample construction We construct the analysis sample in several steps. First, we restrict attention to non-discretionary emergency conditions —such as acute myocardial infarction, stroke, and hip fracture— since they represent high-acuity conditions for which the choice of hospital is effectively non-deferrable and largely outside the patient's control. These emergency conditions are clinically significant and account for a substantial share of inpatient hospital spending among Medicare beneficiaries. We also restrict the estimation sample to emergency hospitalizations arriving via ambulance transport. This ensures that hospital assignment is driven by the ambulance dispatch and routing process rather than by the patient or their family.

To exploit within–ZIP-code geographic variation in exposure to vertically integrated (VI) Medicare Advantage plans, we further restrict the sample to beneficiaries who reside in ZIP codes that span more than one county or are located immediately adjacent to a county boundary. Because MA plan availability is defined at the county level, but beneficiaries' residential ZIP codes often cross county borders, this restriction isolates ZIP codes where individuals are plausibly similar yet face different MA plan menus due to their county assignment. This geographic mismatch generates the key source of within- ZIP variation in VI plan exposure used in the empirical strategy.

Applying these restrictions yields a beneficiary-by–episode-of-care dataset. The unit of observation is an emergency episode for beneficiary i in calendar year t, defined as an inpatient admission originating in the emergency department (or an observation stay subsequently converted to inpatient) with no transfer-in from another acute-care facility. For each episode, we observe the beneficiary's MA plan enrollment at time t, the identity and characteristics of the admitting hospital, and the utilization measure during the inpatient saty.

Our 2019 sample contains approximately 16 thousands emergency episodes meeting the above criteria. In ongoing work, we will expand the dataset to include 2022 Medicare claims, which will significantly increase statistical power and allow us to assess whether patterns of VI hospital effects have changed in the post-COVID period.

6 Empirical Framework

The health plan chosen by enrollee i is denoted $p(i)$ and the hospital where she receives care is $h(i)$. Plan-level vertical integration is $\mathrm{VI}^p_{it} \in \{0,1\}$ with

$$\mathrm{VI}^p_{it} = \begin{cases} 0 & \text{if plan } p(i) \text{ is non-PSHP} \\ 1 & \text{if plan } p(i) \text{ is PSHP} \end{cases}$$

and hospital-level vertical integration is $\mathrm{VI}^h_{it} \in \{0,1\}$ with

$$\mathrm{VI}^h_{it} = \begin{cases} 0 & \text{if hospital } h(i) \text{ is not a plan sponsor} \\ 1 & \text{if hospital } h(i) \text{ is a plan sponsor.} \end{cases}$$

Notice that a beneficiary can be enrolled in a PSHP ($\mathrm{VI}^p_{it} = 1$) and be treated at an integrated hospital ($\mathrm{VI}^h_{it} = 1$) even when the plan and the hospital do *not* belong to the same sponsoring system. Therefore, to capture the "fully integrated dyad" we define a third indicator $\mathrm{VI}^f_{it} \in \{0, 1\}$:

$$\mathrm{VI}^f_{it} = \begin{cases} 0 & \text{if hospital } h(i) \text{ is not the sponsor of plan } p(i) \\ 1 & \text{if hospital } h(i) \text{ is the sponsor of plan } p(i). \end{cases}$$

Baseline specification Let $m(i)$ denote the hospital market (HSA in the main specification) in which the admitting hospital $h(i)$ operates, and let $\widetilde{\ln(HHI)}_{m(i)t}$ denote the market concentration measure (log HHI) *mean-centered* in the estimation sample. Motivating evidence in Section 3 shows that VI hospitals have lower (risk-adjusted) readmission rates in competitive markets, but that the VI–non-VI gap narrows as markets become more concentrated (a positive VI hospital $\times \ln(HHI)$ interaction). To allow the causal VI-hospital effect to vary with market concentration, we estimate:

$$\begin{aligned} y_{it} &= \beta_0 + \beta_1 \mathrm{VI}^p_{it} + \beta_2 \mathrm{VI}^h_{it} + \beta_3 \mathrm{VI}^f_{it} + \beta_4 \widetilde{\ln(HHI)}_{m(i)t} \\ &\quad + \beta_5 \Big(\mathrm{VI}^h_{it} \times \widetilde{\ln(HHI)}_{m(i)t} \Big) + \beta_6 \Big(\mathrm{VI}^f_{it} \times \widetilde{\ln(HHI)}_{m(i)t} \Big) + X'_{it}\Gamma + \alpha_{s(i)} + \delta_{d(i)} + \tau_t + \varepsilon_{it}. \end{aligned} \tag{3}$$

where y_{it} is the 30-day unplanned readmission indicator of individual i in year t. Since $\widetilde{\ln(HHI)}$ is mean-centered, β_2 is the VI-hospital effect evaluated at the average level of concentration, and β_5 captures how the VI-hospital effect noteably attenuates (if $\beta_5 > 0$) as concentration increases.

6.1 Endogeneity and instruments

The integration indicators are endogenous: (i) beneficiaries may select into PSHPs based on unobserved preferences or health risk, and (ii) even in emergencies, unobserved severity and local provider attributes may correlate with being admitted to an integrated hospital. We therefore use an IV strategy with excluded instruments that shift (a) PSHP enrollment and (b) VI-hospital admission.

Ambulance-preference shifter for VI-hospital admission. Following the emergency-care literature, we use an ambulance-company "preference" instrument that captures the propensity of ambulance company a to deliver patients to integrated hospitals. For each ambulance company a, market, year, and emergency condition, we compute the leave-one-out share of transports ending at an integrated hospital among other patients served by that company:

$$\text{AmbShare}_{it} \;=\; \frac{1}{N_{a(i),t,c(i)}-1} \sum_{\substack{j:a(j)=a(i)\\ j\neq i}} \mathbf{1}\{\text{VI}^{h}_{jt}=1\}, \tag{4}$$

where the summation is computed within the same market-year-emergency cell (to preserve relevance and comparability across episodes). Conditional on controls and fixed effects, we assume ambulance assignment affects outcomes only through the admitting hospital (and thus exposure to VI^h), not through unobserved severity or differential pre-hospital treatment.

Differential distance between VI and non-VI hospitals We additionally use a geographic shifter based on relative proximity to integrated vs. non-integrated hospitals. Let Dist_i^{VI} be the distance from beneficiary i's residential ZIP centroid to the nearest integrated (plan-sponsor) hospital, and let Dist_i^{NonVI} be the distance to the nearest non-integrated hospital. Define the *differential distance*:

$$\text{DD}_i \;\equiv\; \text{Dist}_i^{VI} - \text{Dist}_i^{NonVI}. \tag{5}$$

Larger DD_i means the beneficiary lives relatively closer to an integrated hospital (relative to the closest non-integrated alternative). This shifter is relevant for: (i) *PSHP enrollment*, because PSHPs are typically sponsored by local systems and their attractiveness/network convenience increases when enrollees live closer to sponsor hospitals; and (ii) *emergency hospital assignment*, because conditional on ambulance dispatch and routing constraints, proximity mechanically raises the probability that the realized admitting hospital is an integrated one. Our exclusion restriction is that, conditional on X_{it}, diagnosis and time effects, and the sample restrictions (emergency ambulance arrivals; near-border / multi-county ZIPs), DD_i affects y_{it} only through plan and hospital integration exposure, not through unobserved severity or neighborhood-level confounders.

County-level PSHP Availability Index. Medicare Advantage plans' service area is county specific. This implies that beneficiaries who live nearby but residing in different counties might be offered different MA health plans. Our identification strategy relies on the fact that beneficiaries residing in the same zip code might be presented with different choice sets in terms of PSHP. Therefore, we restrict our sample to those zip codes that either span multiple counties, or are located within few kilometers from another county border. We leverage the variation in PSHPs availability across these zip codes as a quasi-exogenous shifter of the probability that a beneficiary enrolls in a PSHP. The identification assumption is that variation in availability of PSHPs in these zip codes is not correlated with within zip code differences in health and demographic characteristics. Let $c(i)$ and $z(i)$ denote the beneficiary i county and zip code respectively. To compare PSHP availability across counties, we compute the number of PSHP per thousand MA beneficiaries. We denote this by "Availability Index", $\text{Avail}_{i,t} \in [0,1]$. The key source of exogenous variation is that beneficiaries who live in the same zip code $z(i)$, but reside in different counties $c(i)$, are exposed to different levels of $\text{Avail}_{i,t}$. [9]

6.2 2SLS implementation with concentration heterogeneity

Define the excluded-instrument vector

$$Z_{it} \equiv \Big(\text{AmbShare}_{it},\ \text{Avail}_{c(i)t},\ \text{DD}_i,\ \ \text{AmbShare}_{it}\times\text{Avail}_{c(i)t}, \text{DD}_i\times\text{Avail}_{c(i)t},\ \text{DD}_i\times\text{AmbShare}_{it}\Big).$$

We treat as endogenous not only $(\text{VI}^p_{it}, \text{VI}^h_{it}, \text{VI}^f_{it})$ but also the interaction terms $\text{VI}^h_{it} \times \ln\widetilde{(HHI)}_{m(i)t}$ and $\text{VI}^f_{it} \times \ln\widetilde{(HHI)}_{m(i)t}$. Following standard practice, we instrument these interaction terms by interacting each excluded instrument with $\ln\widetilde{(HHI)}_{m(i)t}$.

A compact way to write the first stage is:

$$\mathbf{W}_{it} \ = \ \Pi_0 Z_{it} \ + \ \Pi_1\Big(Z_{it} \times \ln\widetilde{(HHI)}_{m(i)t}\Big) \ + \ X'_{it}\Lambda + \alpha_{s(i)} + \delta_{d(i)} + \tau_t + \mathbf{u}_{it}, \tag{6}$$

where

$$\mathbf{W}_{it} \equiv \Big(\text{VI}^p_{it},\ \text{VI}^h_{it},\ \text{VI}^f_{it},\ \text{VI}^h_{it} \times \ln\widetilde{(HHI)}_{m(i)t},\ \text{VI}^f_{it} \times \ln\widetilde{(HHI)}_{m(i)t}\Big)'.$$

The second stage is equation (3), estimated by 2SLS using the instrument set $\{X_{it}, \alpha_{s(i)}, \delta_{d(i)}, \tau_t,\ Z_{it},\ Z_{it}\times \ln\widetilde{(HHI)}_{m(i)t}\}$.

We report Sanderson–Windmeijer first-stage diagnostics for each endogenous regressor. Standard errors are clustered at the ambulance-company level (and, in robustness, two-way clustered at ambulance-company and beneficiary ZIP or county of residence).

[9] We exclude SNPs from both the instrument construction and the analytic sample to keep the menu relevant to non-dual beneficiaries and avoid plan types with distinct eligibility rules.

7 Preliminary Results

The estimation results presented in this section should be considered preliminary. At this stage, we rely only on 2019 data, which limits the available identifying variation—particularly for estimating the effect of full integration, where plan and hospital are jointly vertically integrated. As we incorporate additional years of Medicare Advantage encounters (including 2022), we expect substantially greater identifying variation and statistical power. All results will be updated and refined over the coming weeks.

All outcome models control for beneficiary demographics (age, sex, race), dual eligibility, observable health characteristics (risk score and HCC comorbidities), and plan characteristics. We include principal-diagnosis fixed effects so comparisons are within clinical condition. Our geographic fixed effects are at the state level since our current sample size does not allow us the have more granular fixed effects. Standard errors are clustered at the ambulance-company or health system level. We interpret these results as consistent we the fact that VI plans implement tighter care coordination and stronger post-acute management within integrated systems. This echoes existing evidence that integrated provider–plan structures reduce fragmentation and create incentives to invest in follow-up management.

Table 2 shows the results for 30-day all-cause readmissions. The dependent variable is a binary indicator for whether the beneficiary is readmitted to any hospital within 30 days after her discharge date. Our results indicate that beneficiaries enrolled in a PHSP plan have approximately 33% lower likelihood of being readmitted. The impact of being treated by a VI hospital is not significant. This pattern is consistent with integrated systems' incentives to manage inpatient utilization more aggressively, particularly since VI plans internalize hospital costs.

Table 3: 30-day all cause Readmission

	Odds ratio	Conf. Interval
VI plan	0.669***	[0.502, 0.893]
VI hospital	1.170	[0.935, 1.464]
Observations	16,082	

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Notes: (i) **Sample restriction**: beneficiaries living in zip codes that span more than one county or are located near the county border. (ii) **Dependent variable**: indicator for 30-day all-cause readmission. (iii) principal diagnosis and state FE are included

Table 3 shows the results for the inpatient Length of Stay (LOS). The dependent variable in the logarithm of the day difference between the discharge date and the admission date. Our results indicate that the LOS among beneficiaries enrolled in a PHSP plan is approximately

0.6% shorter than that of beneficiaries enrolled in other plans. The impact of being treated by a VI hospital is not significant.

Table 4: Inpatient Length of Stay (log)

	Coefficient	Robust Std. Error
VI plan	-0.600**	(0.267)
VI hospital	0.089	(0.055)
Observations	16,082	

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Notes: (i) **Sample restriction**: beneficiaries living in zip codes that span more than one county or are located near the county border. (ii) **Dependent variable**: log length of stay for inpatient admissions. (iii) principal diagnosis and state FE are included

Table 4 shows the results for the diagnosis intensity. The dependent variable in the logarithm the number of ICD-10 diagnosis codes. Our results indicate that the diagnosis intensity is lower among beneficiaries enrolled in a PHSP plan by approximately 0.55%, and 0.17% higher among beneficiaries treated by a VI hospital.

Table 5: Diagnosis Intensity (log)

	Coefficient	Robust Std. Error
VI plan	-0.551***	(0.206)
VI hospital	0.171***	(0.042)
Observations	16,082	

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Notes: (i) **Sample restriction**: beneficiaries living in zip codes that span more than one county or are located near the county border. (ii) **Dependent variable**: log number of ICD-10 diagnosis. (iii) principal diagnosis and state FE are included

Table 5 shows the results for the discharge location. The dependent variable is an indicator of whether the beneficiary discharged to another hospital, to a post acute care (PAC) facility or home. The impact of enrolled in a PHSP plan are approximately 65% more likely to be discharge to a PAC facility. The impact of receving care from a VI hospital is not significant.

Taken together, these results indicate that VI plans tend to have lower healthcare utilization in terms of readmission, inpatient length of stay and diagnosis intensity. We think that these patterns are consistent with the fact that vertically integrated MA plans internalize both the cost of hospital care and the downstream financial implications of readmissions, inpatient stay and high-intensity procedures. A second explanation for this higher efficiency is the larger utilization of PAC services.

Table 6: Discharge Location (PAC)

	Odds ratio	Conf. Interval
VI plan	1.652*	[1.057, 2.582]
VI hospital	0.977	[0.794, 1.153]
Observations	16,082	

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Notes: (i) **Sample restriction**: beneficiaries living in zip codes that span more than one county or are located near the county border. (ii) **Dependent variable**: indicator for PAC discharge. (iii) principal diagnosis and state FE are included

8 Conclusions

References

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Appendix A

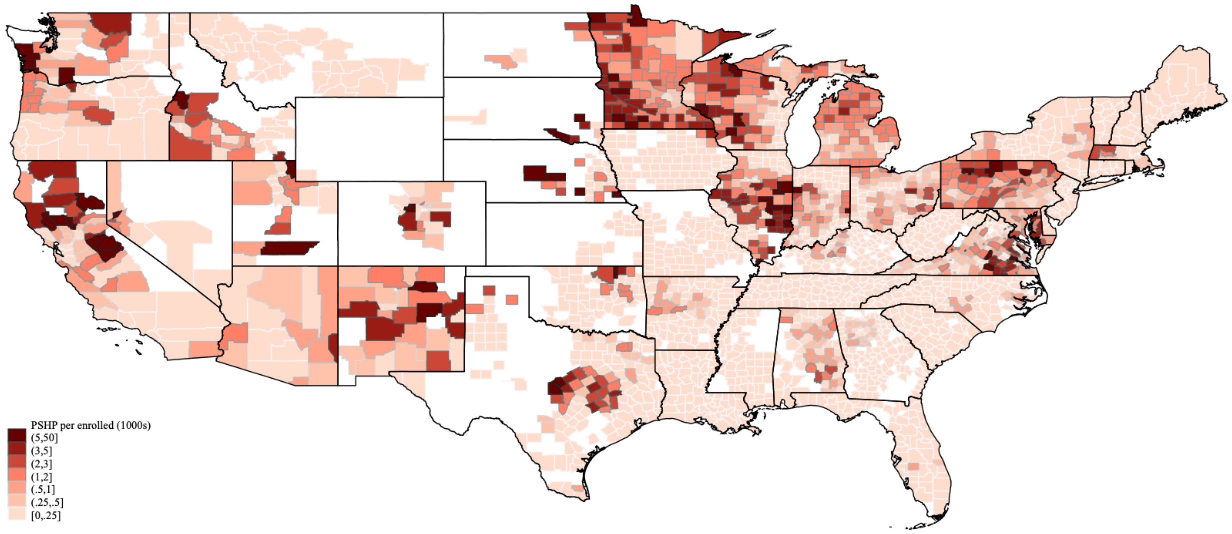

Figure 4: Number of HMO-PSHP per thousands of county HMO enrollees

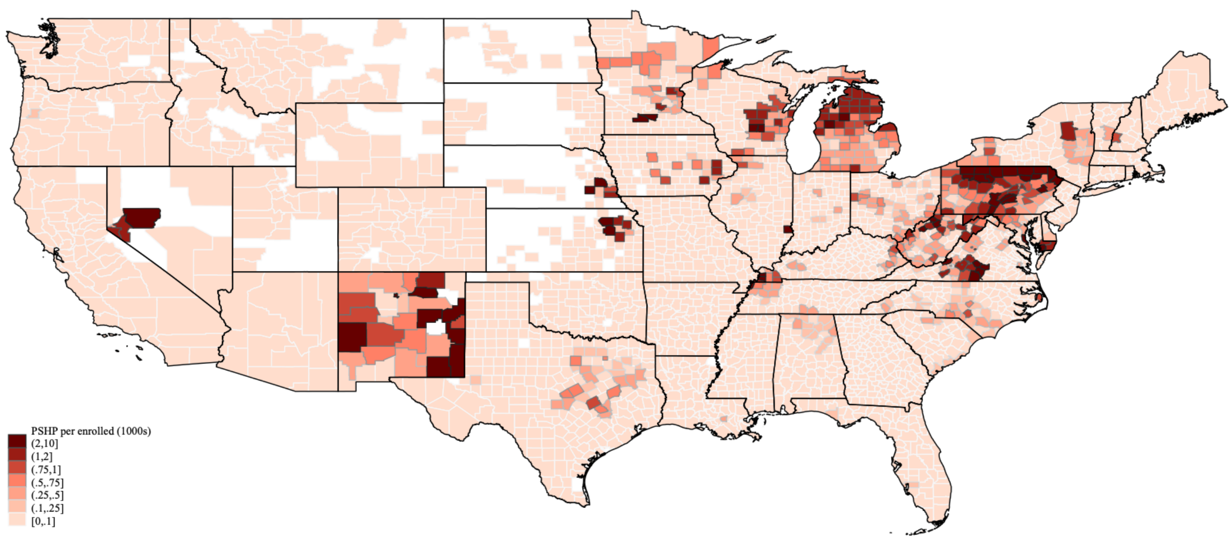

Figure 5: Number of PPO-PSHP per thousands of county PPO enrollees